

Beginning Graphic, Video, and Audio Codecs

Version 1.1; April 2003

Media

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To successfully deliver content, programmers must master not only Multimedia Message Service (MMS), but also the codecs (encoder/decoder) that encode MMS audio and video content. Codecs are used to make audiovisual information more network friendly. In other words, as a developer creates an MMS service, at some point the developer needs to decide which codec to invoke.

This white paper presents the options for Nokia handset developers and presents the basic information needed regarding which codecs are supported and how they should be used in MMS applications.

While they may look daunting to those who have not worked with codecs before, the idea behind them is simple. A codec is a program, together with its associated data format, that is used for compressing and decompressing digital audio and graphical information so that it can be sent quickly over the available bandwidth and be easily stored and displayed/played back by the receiver. The terms “coding” and “decoding” are also used.

1. Codec Overview

Most users use codecs every day on the Web, often without realizing it, with graphical formats like Joint Photographic Experts Group (JPEG) images and RealAudio music files. Like the Web’s HTTP protocol, on one level, MMS can be thought of as part of the transport level for encoded data.

MMS is more than just a transport level, however. It is also a store-and-forward protocol that enables mobile phone users to retrieve messages when they desire. MMS messages are sent and then stored on a Multimedia Message Service Center (MMSC). After the message arrives, the MMSC sends the receiver’s phone a Short Message Service (SMS) notification message via WAP Push. The MMS client in the receiving terminal then can either automatically or manually, depending on the user’s message settings, retrieve MMS messages with a WAP GET call.

If the user cannot read MMS messages because he or she is using an older, non-MMS-capable device or a device without enough memory to receive and open the message, a Web URL will be sent to enable the recipient to view the MMS message with a Web browser. That may not be necessary, however, even if the user’s terminal is not capable of correctly displaying a particular message. When the recipient is fetching the message, the MMSC checks the user agent header, recognizes the capabilities of the receiving terminal, and converts the content to be suitable for the recipient.

For example, the MMSC can do media-type conversion, such as turning video into a slide show. Or, for instance, media format conversion can be made by changing Graphics Interchange Format (GIF) to JPEG images, or by changing a picture’s size or resolution without the MMS programmer having to worry about writing code that results in reducing all the media types to the least common denominator. The MMSC will do most of the conversion work. However, some phones, such as the Nokia 3510i, do not have Synchronized Multimedia Integration Language (SMIL) support and, therefore, are not capable of showing slides.

MMS also defines which of the dozens of available codecs can be used in its messages. The mandatory codecs are described in the *MMS 2.0 Conformance* document (http://www.openmobilealliance.org/omacopyrightNEW.asp?doc=OMA-MMS-v1_1-20021104-C.zip), such as baseline JPEG and Adaptive Multi-Rate (AMR). The total number of mandatory codecs, counting text codecs, is 10. However, developers only need to use the codecs that are necessary to support their own applications. For example, a slide

show program that uses images, accompanied by music produced by Musical Instrument Digital Interface (MIDI) audio does not need to support a video codec, since the application does not send video as part of the MMS message.

Each valid codec file format, or MMContent content-type as it is known in MMS programming terminology, is defined as a MIME type. For more on MIME, see MIME (<http://www.wapforum.org/WINA/wsp-content-type.htm>). All MMS messages will require the appropriate MMContent content-types set so that the mobile device and the MMSC can handle them appropriately.

2. Working with Other Formats

Series 60 devices, including the Nokia 3650 and Nokia 7650, can also receive and play back other formats when a customized application — like the RealOne player — can cope with them. Such an application must “announce” its MIME type and provide a “recognizer” to map a file to a particular MIME type. A recognizer does this by examining both the file extension and the actual file contents — usually the file’s first few bytes.

In using such methods, though, developers will no longer be using MMS. Instead, they will be building an application on top of the Nokia device’s operating system, in this case Symbian OS. This is an alternative path to providing users with multimedia presentations and is not recommended for most developers.

3. The SMIL Envelope

MMS messages are sent using SMIL as the presentation language. On Series 30 and 40 phones, however, MMS uses SMIL but does not display SMIL messages according to either SMIL 2 or SMIL Basic. This is because SMIL was designed for use with PC terminals and not on the much smaller mobile phone displays of those devices. Therefore, Series 30 and 40 phones will ignore SMIL’s layout section. Series 60 devices will show the MMS message according to the SMIL instructions, but some of the parameters are ignored, such as <dur> (duration of one slide).

On Series 30 and 40 handsets, the media elements of an MMS message will be displayed in the order in which they are sent, not in the order specified by SMIL. As a result, when a user sends an SMIL message with audio, the text and image portions must be saved separately, and then reconstructed into a new MMS message in order to forward it. In those cases, the original SMIL presentation component is lost.

No matter which MMS-capable phone is the target terminal, SMIL must still be used to format MMS messages. But in the case of audio and image or video sent to Series 30 and 40 phones, the SMIL instructions will be ignored by the MMSC. In those cases, the actual presentation will be determined by the order in which the sender sent the elements of an MMS message, and not per the SMIL instructions.

In the future, SMIL will enable developers to create MMS multimedia presentation programs that will let users describe more precisely how the multimedia parts of a given message should be displayed. MMS authoring tools using SMIL eventually should become very popular with users.

4. The Codecs

No matter which codecs are used, MMS does have some practical limits. Currently, MMS messages can be no larger than 100 KB. Thus, regardless of the codec, and regardless of bandwidth or device memory, there are still practical limits to total file size. Unlike SMS, with its 160-character limit, this is not a hard-wired limitation. At this early stage of development, 100 KB is the practical limit. As the technology improves and bandwidth increases, MMS messages will be able to grow to meet the demands of users.

As described previously, MMS codecs use MIME formats. This means that the general media type is called out first, e.g., “audio” for sound and “image” for graphics. The general media type is then followed by the specific codec or program for a given MIME type, so, for example, a GIF image is described as image/gif.

Audio: Many Series 40 and some Series 30 Nokia handsets support MIDI (audio/midi) audio with up to 43 instruments possible and up to four of them playing at one time. Series 60 phones such as the Nokia 3650 and Nokia 7650 support up to 24 voices playing simultaneously. This allows a user to use the Star Wars theme music, for example, for polyphonic ring tones.

Images: Anyone who has programmed for the Web will not be surprised that JPEG (image/jpeg), GIF87a, and GIF89a, including animated GIF89a (image/gif), as well as wireless Bitmap (WBMP, image/vnd.wap.wbmp) formats are mandatory MMS formats. In addition, Nokia supports the optional Portable Network Graphics (PNG, image/png) standard in its devices.

Regardless of which image format is used, developers should keep in mind that the maximum guaranteed MMS resolution is 160 x 120 pixels. In practice, Series 60 handsets, like the Nokia 7650, can display up to 128 x 115 pixels in the MMS viewer out of a display size of 160 x 128 pixels.

Personal Information Management (PIM): Nokia supports both mandatory MMS formats: vCard 2.1 (text/x-vCard) and vCalendar 1.0 (text/x-vCalendar). Nokia does not support the optional “vCard 2.0 plus picture” format. However, developers can use the Internet Mail Consortium’s vCard and vCalendar to create both business card exchange and scheduling programs and links to existing PC-based organizers built around Microsoft Exchange and Lotus Notes, as well as others. For more information on these two formats, see the Personal Data Interchange site (<http://www.imc.org/pdi>). Technically, none of the vCards are codecs, but they are treated as such in the *MMS 2.0 Conformance* document, hence their inclusion here.

Plain Text: In MMS, plain text is taken from Unicode’s Latin character sets (charset). The supported text formats are US-ASCII, UTF-8, and UTF-16 with explicit Byte Order Mark (BOM). This means, in theory, that MMS can also be used as a replacement for SMS for text messaging.

Practically speaking, text in MMS is used as notes for multimedia content. To use text as an MMContent content-type, it can be defined as: “Content-type: text/plain; charset=US ASCII”. Without a character set specification, MMS’s text/plain will default to UTF-16 BOM. On the Internet, however, the default MIME text character set is US-ASCII, so developers will need to be careful about simply taking ordinary Web text and inserting it into an MMS message.

For more about Unicode, see the Unicode Home Page (<http://www.unicode.org/>).

Voice: Adaptive Multi-Rate (AMR) is the mandatory voice codec for third-generation (3G) cellular systems. When combined with H.263 video, AMR can also be used to create bandwidth-effective “3GP” multimedia clips. 3GP is used in the Nokia Video Player and is integrated with both the e-mail and MMS client with a maximum bitstream of 64 kilobit/s (Kbit/s).

AMR’s full technical specification can be found at the 3rd-Generation Partnership Project (3GPP) site (http://www.3gpp.org/ftp/tsg_sa/TSG_SA/TSGS_04/Docs/PDF/SP-99245.pdf).

Video: The Nokia Video Player supports the International Telecommunications Union standard H.263 codec, which was designed for video coding at low bit rates, beginning with 20 to 30 kilobits per second (Kbps) (thousand bits per second) and moving up to 64 Kbps.

In MIME terms, such files or broadcasts are sent in two files with two formats: audio/3gpp and video/3gpp. These are then integrated together for the viewer. Nokia supports the 3GP MMS video codec.

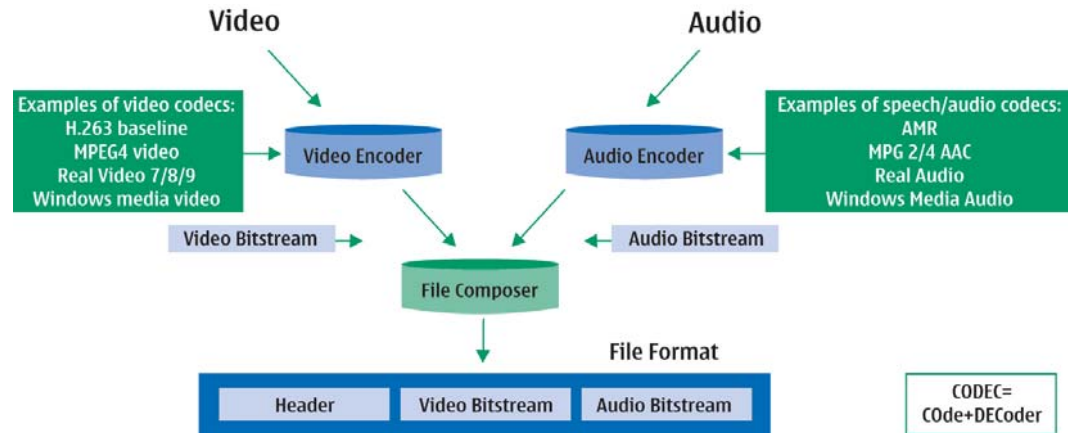


Figure 1: An example of codecs being used to build a video clip

Developers can also use RealNetworks's RealOne Player for Mobile Devices, available for download from http://www.realnetworks.com/mobile/nokia/index.html?src=120302realhome_1, to view the following RealAudio and RealVideo-supported formats: audio/x-pn-realaudio; application/x-pn-realmedia; application/vnd.rn-realmedia; audio/vnd.rn-realaudio; and video/vnd.rn-realvideo; as well as video/3gpp and audio/3gpp formats. The player is bundled with some mobile devices, such as the Nokia 3650, or can be downloaded from the Real Web site.

In addition, PacketVideo Mobilemedia's (<http://www.packetvideo.com/>) pvPlayer and pvAuthor on Series 60 phones from Nokia can also download or stream video and audio clips in Motion Picture Experts Group (MPEG-4) format.

Regardless of which format is used, developers should always keep in mind that video file sizes in MMS messages are currently limited to a maximum of 100 KB.

5. Beyond the Codec

To begin deploying codecs, programmers should start with *How to Create MMS Services*. Next, take a close look at *Video in Nokia Phones* for the essentials on using 3GP and Real formats in MMS. Because MIDI, with its small size and large user support, is also likely to be popular, developers should also read *MIDI in Nokia Phones*. All these documents can be found at www.forum.nokia.com/main/1,6566,21,00.html.

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